Roll Number: _____ Name_____ Group_____

THAPAR INSTITUTE OF ENGINEERING &TECHNOLOGY, PATIALA
Department of Computer Science & Engineering
Theory of Computation (UCS701) – End Semester Examination
Date: 27/05/2025 & Time: 2:00PM MM: 60 & MT: 180 Min

List of Faculties: Dr. Chinmaya Panigrahy, Dr. Shashank Sheshar Singh, Dr. Rohan Sharma, Dr. Aditi Sharma, Dr. Vibha Jain, Dr. Nidhi Chakravarty, Dr. G.V. Eswara Rao, and Mr. Abdul Kadir

Attempt any three of the four questions. Attempt/Answer all sub-parts like (a), (b), (c) for each question in one place. Do mention Page No. of your attempt on the front page. Question paper is of two pages. Please turn over this page to see the questions on the next page.

Ques. No		Questions	Marks	CO	BL
Q1.	(a)	Consider the language, L = {wXw^R | w ∈ (a+b)*. Here, w^R represents the reverse of w. Here, X is another alphabet. For the L, (i) Draw a flow diagram for the corresponding PDA. (ii) Design a PDA to accept L. (iii) What is the tuple representation of the PDA constructed in (ii)? (iv) Show the sequence of ID's the PDA will move through while processing the string abbXbba.	10	CO2	L3
	(b)	Show that the language, L = {a$^{n!}$ | n ≥ 0} is not context-free using the Pumping lemma for context-free languages.	3	CO2	L3
	(c)	Use the CYK algorithm to determine whether the string bbaba is in the language generated by the following grammar. S → AB | BC A → BB | a B → BA | b C → AC | AA | a	5	CO2	L3
	(d)	Prove that the context-free languages are closed under Kleene closure operation.	2	CO2	L3
Q2.	(a)	Design a Turing machine for the language, L = {0^X | X=n^2, where n ≥ 0 }, by writing the logic. Show the processing of the string, w = 000 by the constructed Turing machine.	8	CO3	L4
	(b)	Write down the logic for designing a Turing machine over {a, b} for the language, L={ww | w ∈ (a+b)*}.	4	CO3	L4
	(c)	Design a multi-tape Turing machine to accept the language, L = {a^{n}b^{n}c^n | n > 0}.	4	CO3	L4
	(d)	Design a Moore machine to find the occurrence of aba in the input string over Σ = {a,b}. Convert the above Moore machine to an equivalent Mealy machine. Show the outputs of the above machines for the input, w = ababa.	4	CO1	L3
Q3.	(a)	Convert the following context-free grammar into Greibach normal form. S → XA | BB B → SB | b X → b A → a	7	CO2	L4
	(b)	Prove that the language, L = {a^{n}b^{n}c^{m}d^m ∪ a^{n}b^{m}c^{m}d^n | n, m ≥ 1} is inherently ambiguous.	5	CO2	L4

	(c)	Simplify the following CFG. $S \rightarrow aA \mid aBB$ $A \rightarrow aAA \mid \varepsilon$ $B \rightarrow bB \mid bbC$ $C \rightarrow B$	**2**	**CO2**	**L3**
	(d)	Draw Finite Automata for the following languages: L1={x | 00 is not substring of x, x ∈ {0,1}*} L2={x | x ends with 01, x ∈ {0,1}*} Then use the finite automata for L1 and L2 and construct the finite automata for L1 ∪ L2, L1 ∩ L2 and L1 − L2.	**6**	**CO1**	**L4**
Q4.	**(a)**	Construct a right-linear grammar for the regular expression, 110(01+001)*011. Then, construct a finite automata from the resultant right-linear grammar.	**5**	**CO1**	**L4**
	(b)	A string y is said to be a permutation of the string x if the symbols of y can be reordered to make x. For instance, the permutations of string x = 011 are 110, 101, and 011. If L is a language, then perm(L) is the set of strings that are permutations of strings in L. For example, if L= $\{0^n1^n \mid n>=0\}$, then perm(L) is the set of strings with equal numbers of 0's and 1's. (i) Give an example of a regular language L over alphabet {0, 1} such that perm(L) is not regular. Justify your answer. Hint Try to find a regular language whose permutations are all strings with an equal number of 0's and 1's. (ii) Give an example of a regular language L over alphabet {0, 1, 2} such that perm(L) is not context-free. (iii) Prove that for every regular language L over a two symbol alphabet, perm(L) is context-free.	**10**	**CO4**	**L4**
	(c)	Consider a language L which consists the set of strings on {0,1} in which, every substring of 3 symbols has at most two zeros. For example, 001110 and 011001 are in the language, but 100010 is not. All strings of length less than 3 are also in the language. Construct the minimal DFA for language L.	**5**	**CO1**	**L3**

.....All the Best.....